MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW POWER SCHOTTKY TTL, 4-BY-4 REGISTER FILE, CASCADABLE, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Schottky TTL, low power, 4-by-4 register file microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	4-by-4 register file with 3-state outputs, cascadable
02	4-by-4 register file with open-collector outputs cascadable

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation per register (P_D) 1/	
Device 01	275 mW dc
Device 02	220 mW dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC})	
Cases E and F	(see MIL-M-38510, appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).

2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage range (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.7 V dc
Case operating temperature range (T_C)	-55°C to +125°C
Minimum width of write-enable or read-enable pulse	25 ns
Minimum setup time (data)	10 ns
Minimum setup time (write-select)	15 ns
Minimum hold time (data)	15 ns (With respect to GW)
Minimum hold time (write-select)	5 ns
Minimum latch time for new data	25 ns

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.2 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.3 Truth tables. The truth tables shall be as specified on figure 2.

3.2.4 Logic diagrams. The logic diagrams shall be as specified on figure 3.

3.2.5 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained by the agent activity and will be available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

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3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics of table I apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspections. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 5 or equivalent.

(2) T_A = +125°C.

b. Interim and final electrical test parameters shall be as specified in table II; the interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 5 or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration = 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 2.0 V, I_{OH} = -1 mA	01	2.4		V
Collector cutoff current	I_{CEX}	V_{CC} = 4.5 V, V_{OH} = 5.5 V	02		20	µA
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 4 mA V_{IN} = 0.7 V	01,02		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C	01, 02		-1.5	V
High level input current, all inputs except GR and GW	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01, 02		20	µA
High level input current, at GR	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01		60	µA
			02		40	µA
High level input current, at GW	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01, 02		40	µA
High level input current, all inputs except GR and GW	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01, 02		100	µA
High level input current, at GR	I_{IH5}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01		300	µA
			02		200	µA
High level input current, at GW	I_{IH6}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01, 02		200	µA
Off state output current, high level voltage applied	I_{OZH}	V_{CC} = 5.5 V, V_O = 2.7 V	01		20	µA
Off-state output current, low level voltage applied	I_{OZL}	V_{CC} = 5.5 V, V_O = 0.4 V	01		-20	µA
Low level input current at data, read select, and write select	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01, 02	-0.5	-460	µA

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Low level input current at read enable	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01	-90	-1300	µA
			02	-90	-900	µA
Low level input current at write enable	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01, 02	-60	-840	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 1/	01	-15	-130	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01		50	mA
			02		40	mA
Propagation delay time, low-to-high level from data	t_{PLH1}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 316 Ω ±5% for device type 01; R_L = 2 kΩ ±5% for device type 02	01	2	65	ns
			02	2	65	ns
Propagation delay time, low-to-high level from read select	t_{PLH2}		01	2	58	ns
			02	2	58	ns
Propagation delay time, high-to-low level from data	t_{PHL1}		01	2	58	ns
			02	2	52	ns
Propagation delay time, high-to-low level from read select	t_{PHL2}		01	2	65	ns
			02	2	58	ns
Propagation delay time, low-to-high level from write enable	t_{PLH3}		01	2	65	ns
			02	2	65	ns
Propagation delay time, low-to-high level from read enable	t_{PLH4}		02	2	46	ns
Propagation delay time, high-to-low level from write enable	t_{PHL3}		01	2	72	ns
			02	2	58	ns
Propagation delay time, high-to-low level from read enable	t_{PHL4}		02	2	46	ns
Output-enable time to low level	t_{PZL}		01	2	58	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Unit
Output enable time to high level	t_{PZH}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 316Ω ±5% for device type 01; R_L = 2 kΩ ±5% for device type 02	01	2	52	ns
Output disable time to low level	t_{PLZ}		01	2	52	ns
Output disable time to high level	t_{PHZ}		01	2	72	ns

1/ Not more than one output should be shorted at a time.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) Class S devices	Subgroups (see table III) Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9,10,11
Group B test requirements (method 5005)	1,2,3,7,8,9,10,11	---
Group C end-point electrical parameters (method 5005)	---	1,2,3
Additional electrical subgroups for group C periodic inspections	---	None
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

* PDA applies to subgroup 1 (see 4.2c).

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

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Device types 01 and 02

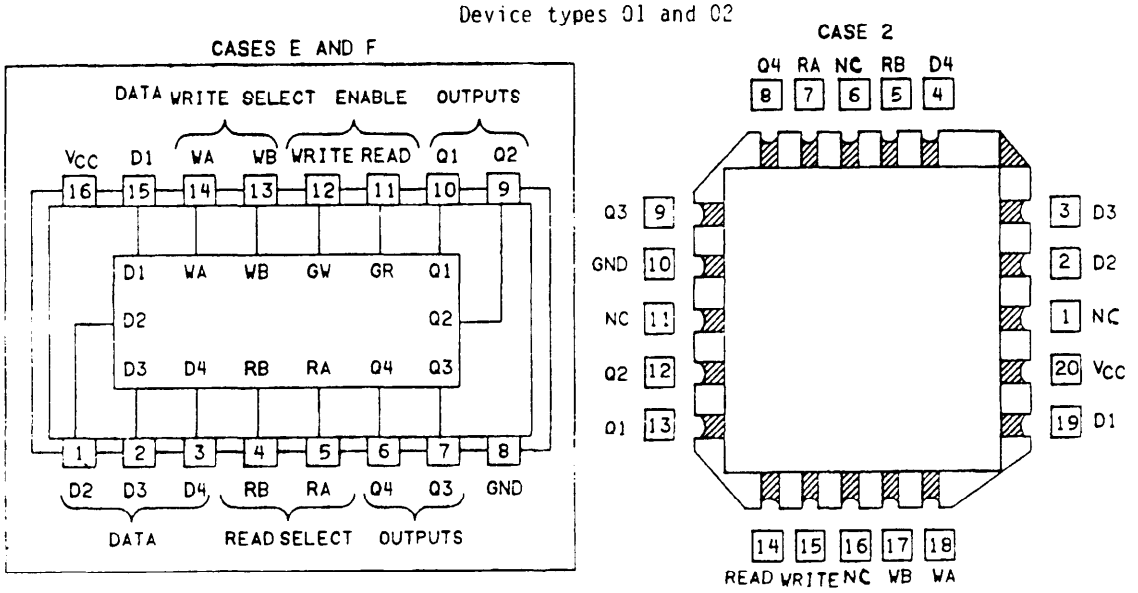


FIGURE 1. Terminal connections. (TOP VIEW)

Device type 01

Write function table (see notes A, B, and C)

Write inputs			Word			
WB	WA	GW	Q	1	2	3
L	L	L	Q = D	Q_0	Q_0	Q_0
L	H	L	Q_0	Q = D	Q_0	Q_0
H	L	L	Q_0	Q_0	Q = D	Q_0
H	H	L	Q_0	Q_0	Q_0	Q = D
X	X	H	Q_0	Q_0	Q_0	Q_0

Read function table (see notes A and D)

Read inputs			Outputs			
RB	RA	GR	Q1	Q2	Q3	Q4
L	L	L	WOB1	WOB2	WOB3	WOB4
L	H	L	W1B1	W1B2	W1B3	W1B4
H	L	L	W2B1	W2B2	W2B3	W2B4
H	H	L	W3B1	W3B2	W3B3	W3B4
X	X	H	Z	Z	Z	Z

NOTES:

A. H = high level, L = low level, X = irrelevant, Z = high impedance (off)
B. (Q = D) - The four selected internal flip-flop outputs will assume the states applied to the four external data inputs.
C. Q_0 = the level of Q before the indicated input conditions were established.
D. W0B1 = The first bit of word 0, etc.

Device type 02

Write function table (see notes A, B, and C)

Write inputs			Word			
WB	WA	GW	Q	1	2	3
L	L	L	Q = D	Q_0	Q_0	Q_0
L	H	L	Q_0	Q = D	Q_0	Q_0
H	L	L	Q_0	Q_0	Q = D	Q_0
H	H	L	Q_0	Q_0	Q_0	Q = D
X	X	H	Q_0	Q_0	Q_0	Q_0

Read function table (see notes A and D)

Read inputs			Outputs			
RB	RA	GR	Q1	Q2	Q3	Q4
L	L	L	WOB1	WOB2	WOB3	WOB4
L	H	L	W1B1	W1B2	W1B3	W1B4
H	L	L	W2B1	W2B2	W2B3	W2B4
H	H	L	W3B1	W3B2	W3B3	W3B4
X	X	H	H	H	H	H

NOTES:

A. H = high level, L = low level, X = irrelevant.
B. (Q = D) = The four selected internal flip flop outputs will assume the states applied to the four external data inputs.
C. Q_0 = the level of Q before the indicated input conditions were established.
D. W0B1 = The first bit of word 0, etc.

FIGURE 2. Truth tables.

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Device type 01

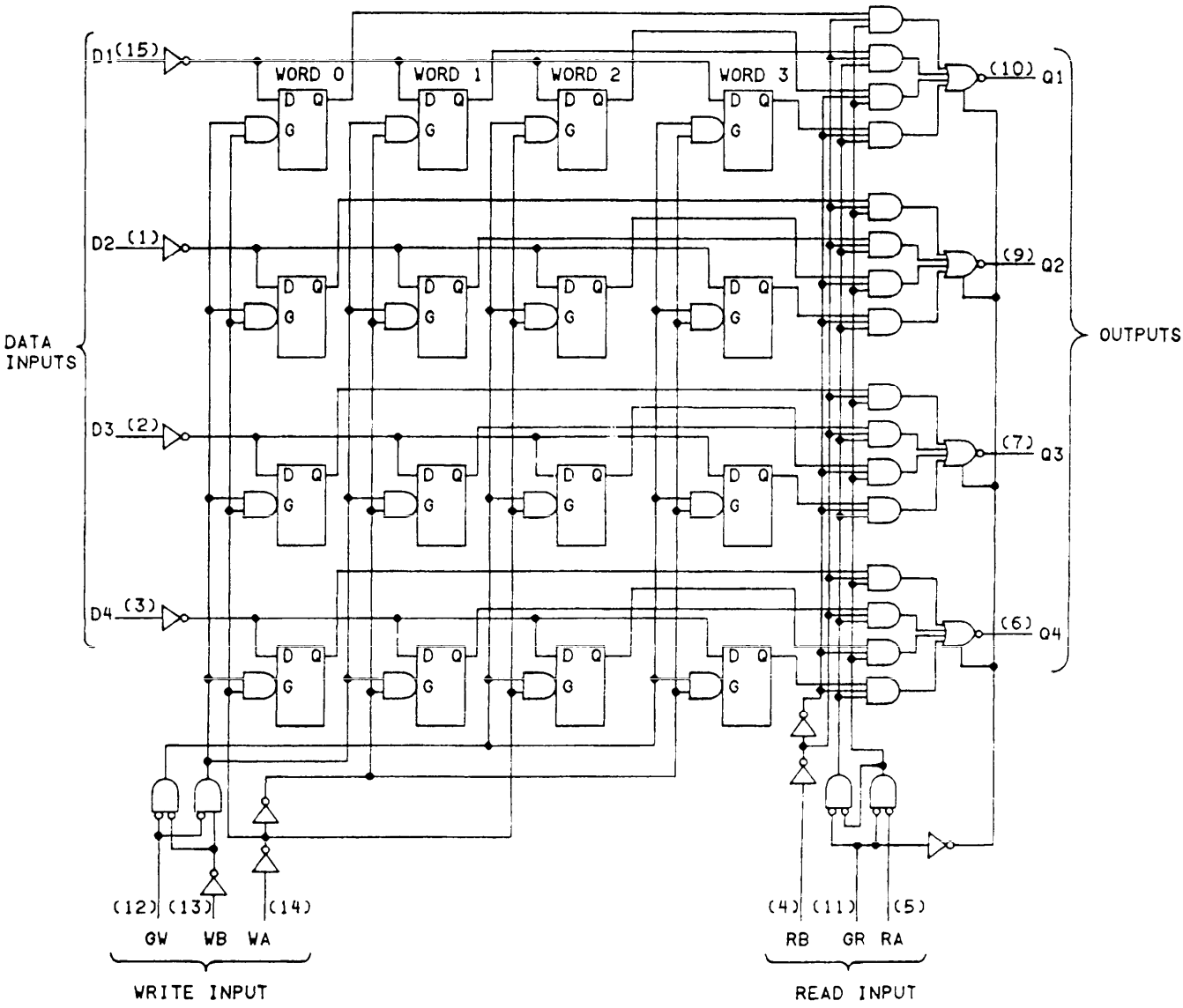


FIGURE 3. Logic diagrams.

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Device type 02

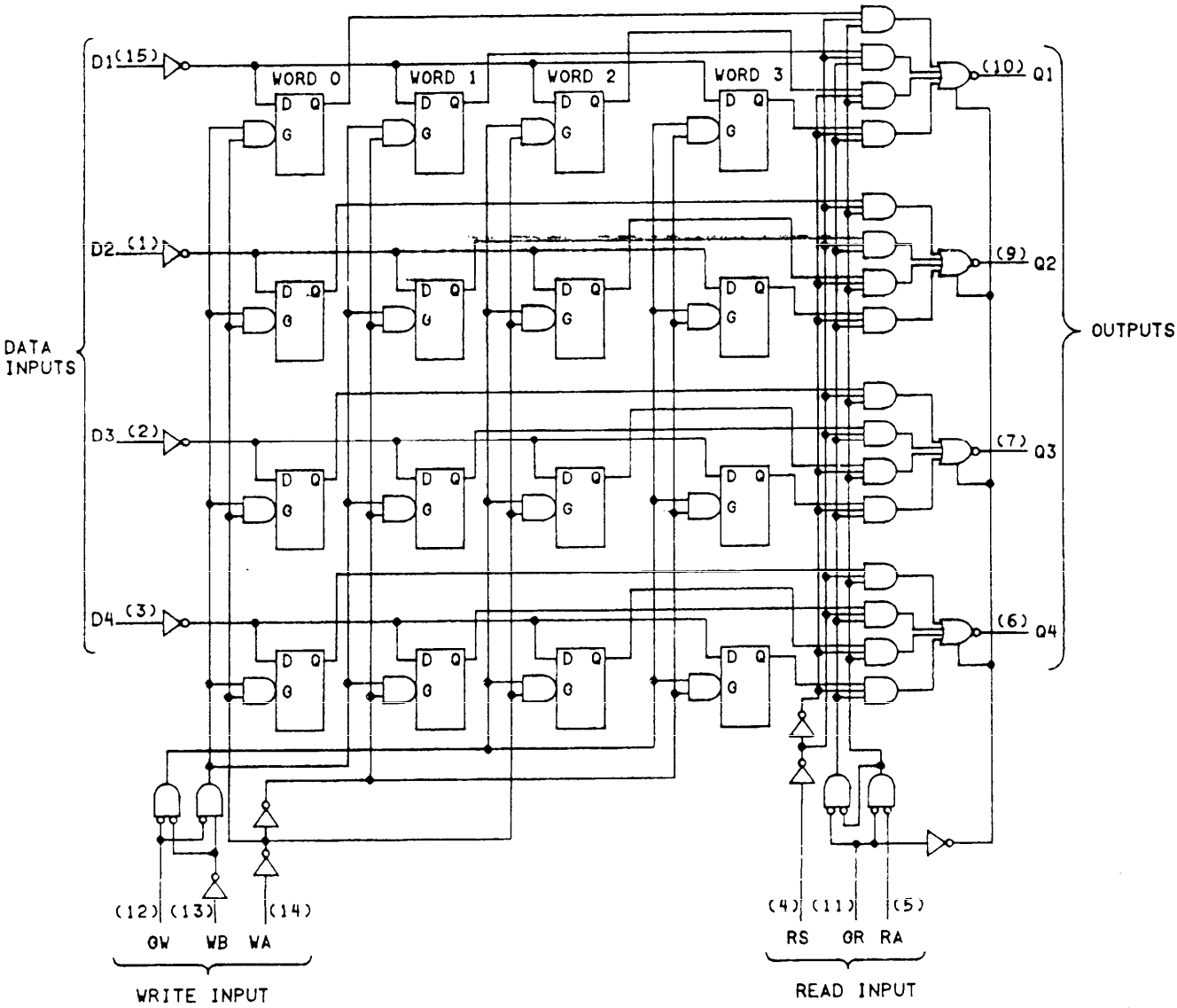
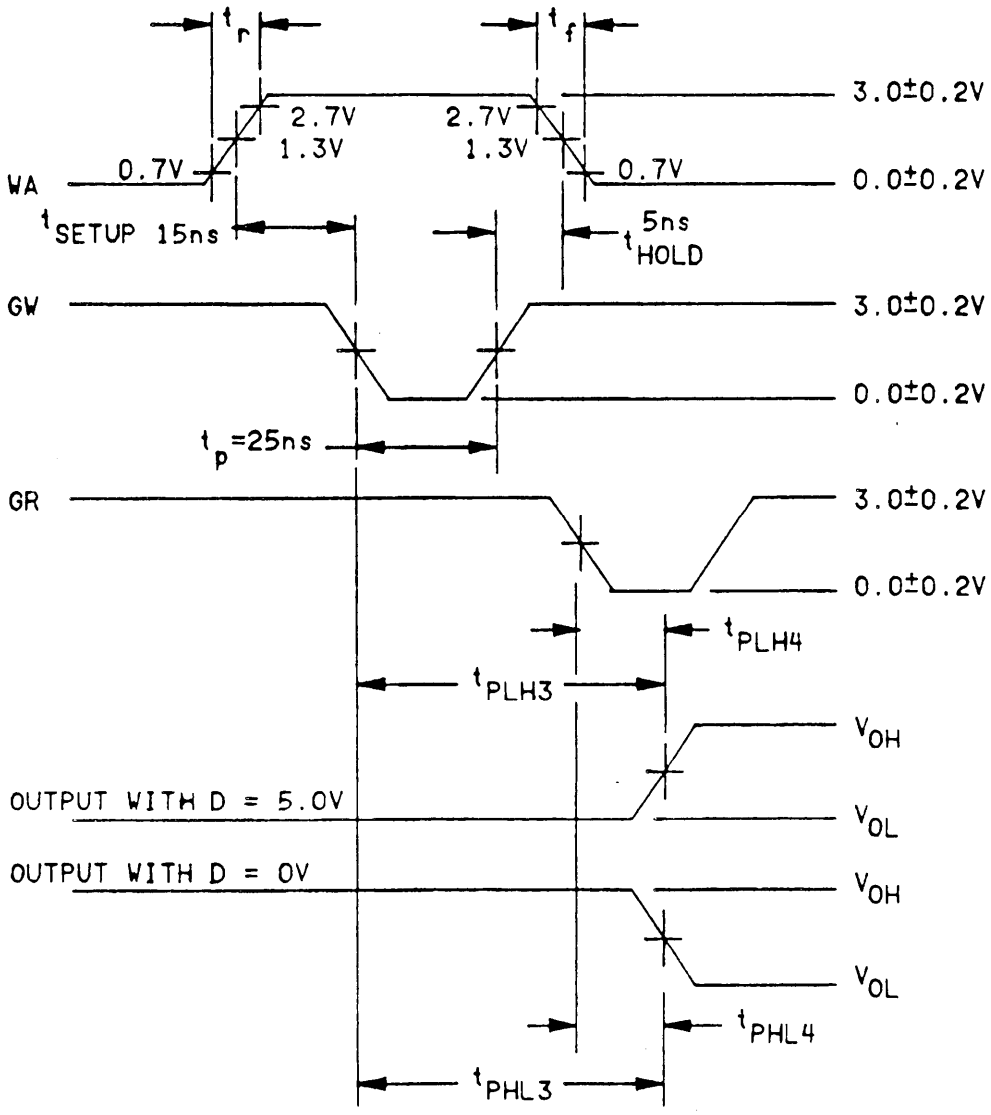


FIGURE 3. Logic diagrams - Continued.

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NOTES:
1. Input pulse characteristics: PRR $\leq$ 1.0 MHz, $t_r \leq$ 15 ns, and $t_f \leq$ 6 ns; duty cycle = 50% ±15%.
2. C_L = 50 pF ±10%.
   C_L includes probe and jig capacitance.
3. All diodes are 1N3064 and 1N916.
4. Load circuits on a given output are required only where the specific test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 4. Switching test curcuit and waveforms, device types 01 and 02.

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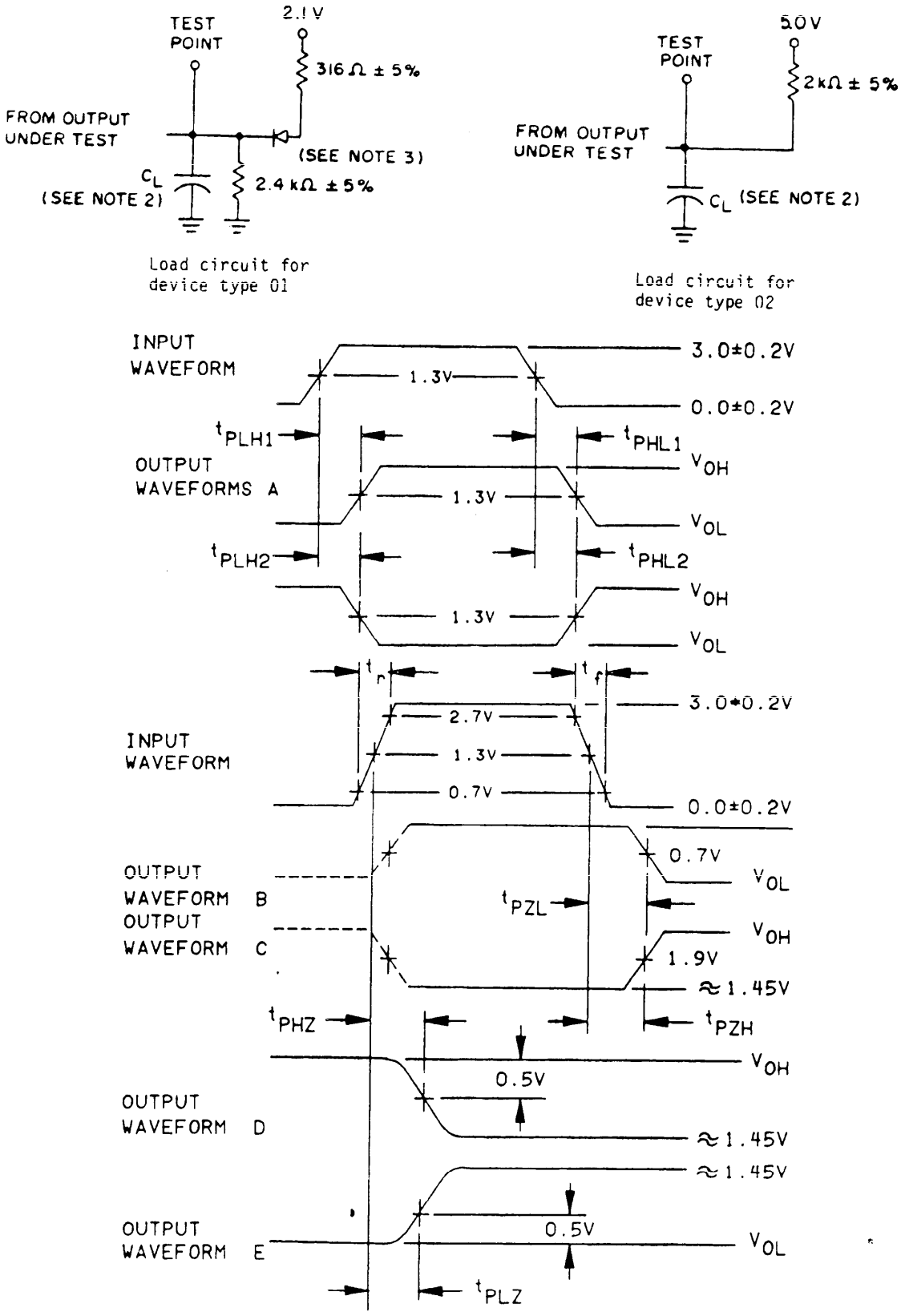
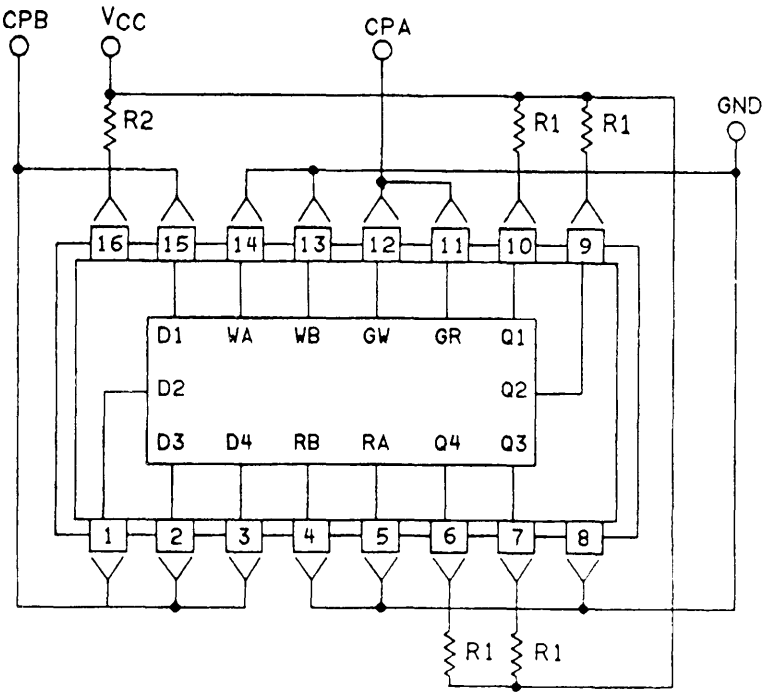


FIGURE 4. Switching test circuit and waveforms, device types 01 and 02 - Continued.

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NOTES:
1. CPA = 100 kHz ±10% square wave; duty cycle = 50%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. CPB = 50 kHz ±10% square wave synchronized with CPA. All other conditions are the same as CPA.
3. Output pins, R_1 = 1.0 kΩ ±10%; input and V_{CC} pins, R_2 = 27Ω ±10%.
4. V_{CC} shall be high enough to ensure 5.0 V minimum at the device terminal.

FIGURE 5. Burn-in and life test circuit, types 01 and 02.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be High ≥2.0 V, low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2 Cases E,F Test no.	2 1 D2	3 2 D3	4 3 D4	5 4 RB	7 5 RA	8 6 Q4	9 7 Q3	10 8 GND	12 9 Q2	13 10 Q1	14 11 GR	15 12 GW	17 13 WB	18 14 WA	19 15 D1	20 16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1				0.7 V	0.7 V			GND		4 mA	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	4.5 V	Q1		0.4	V
		"	2	0.7 V			"	"			"	4 mA		"	"	"	"		"	Q2		"	"
		"	3		0.7 V		"	"		4 mA	"			"	"	"			"	Q3		"	"
		"	4			0.7 V	"	"	4 mA		"			"	"	"			"	Q4		"	"
	V_{OH}	3006	5				"	"			"		-1 mA	"	"	"	"	2.0 V	"	Q1	2.4		"
		"	6	2.0 V			"	"			"	-1 mA		"	"	"	"		"	Q2	"		"
		"	7		2.0 V		"	"		-1 mA	"			"	"	"	"		"	Q3	"		"
		"	8			2.0 V	"	"	-1 mA		"			"	"	"	"		"	Q4	"		"
	V_{IC}		9	-18 mA							"								"	D2		-1.5	"
			10		-18 mA						"								"	D3		"	"
			11			-18 mA					"								"	D4		"	"
			12				-18 mA				"								"	RB		"	"
			13					-18 mA			"								"	RA		"	"
			14								"			-18 mA					"	GR		"	"
			15								"				-18 mA				"	GW		"	"
			16								"					-18 mA			"	WB		"	"
			17								"						-18 mA		"	WA		"	"
			18								"							-18 mA	"	D1		"	"
	I_{IL1}	3009	19	0.4 V							"								5.5 V	D2	1/	1/	µA
		"	20		0.4 V						"								"	D3	"	"	"
		"	21			0.4 V					"								"	D4	"	"	"
		"	22				0.4 V				"								"	RB	"	"	"
		"	23					0.4 V			"								"	RA	"	"	"
		"	24								"					0.4 V			"	WB	"	"	"
		"	25								"						0.4 V		"	WA	"	"	"
		"	26								"							0.4 V	"	D1	"	"	"
	I_{IL2}	"	27								"			0.4 V					"	GR	"	"	"
	I_{IL3}	"	28								"				0.4 V				"	GW	"	"	"
	I_{IH1}	3010	29	2.7 V							"								"	D2		20	"
		"	30		2.7 V						"								"	D3		"	"
		"	31			2.7 V					"								"	D4		"	"
		"	32				2.7 V				"								"	RB		"	"
		"	33					2.7 V			"								"	RA		"	"
		"	34								"					2.7 V			"	WB		"	"
		"	35								"						2.7 V		"	WA		"	"
		"	36								"							2.7 V	"	D1		"	"
	I_{IH2}	"	37								"			2.7 V					"	GR		60	"
	I_{IH3}	"	38								"				2.7 V				"	GW		40	"
	I_{IH4}	"	39	5.5 V							"								"	D2		1100	"
		"	40		5.5 V						"								"	D3		"	"
		"	41			5.5 V					"								"	D4		"	"
		"	42				5.5 V				"								"	RB		"	"
		"	43					5.5 V			"								"	RA		"	"
		"	44								"					5.5 V			"	WB		"	"
		"	45								"						5.5 V		"	WA		"	"
		"	46								"							5.5 V	"	D1		"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2 Cases E,F Test no.	2 1 D2	3 2 D3	4 3 D4	5 4 RB	7 5 RA	8 6 Q4	9 7 Q3	10 8 GND	12 9 Q2	13 10 Q1	14 11 GR	15 12 GW	17 13 WB	18 14 WA	19 15 D1	20 16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IH5}	3010	47								GND			5.5 V					5.5 V	GR		300	µA
	I_{IH6}	"	48								"				5.5 V				"	GW		200	"
	I_{CC}	3005	49	4.5 V	4.5 V	4.5 V	GND	GND			"			4.5 V	4.5 V	GND	GND	4.5 V	"	V_{CC}		50	mA
	I_{OS} 2/	3011	50				"	"			"		GND	GND	GND	"	"	5.5 V	"	01	-30	-130	"
		"	51	5.5 V			"	"			"	GND		"	"	"	"	"	"	02	"	"	"
		"	52		5.5 V		"	"		GND	"			"	"	"	"	"	"	03	"	"	"
		"	53			5.5 V	"	"	GND		"			"	"	"	"	"	"	04	"	"	"
	I_{OZH}		54				0.7 V	0.7 V			"		2.7 V	2.0 V	0.7 V	0.7 V	0.7 V	2.0 V	"	01		20	µA
			55	2.0 V			"	"			"	2.7 V		"	"	"	"		"	02		"	"
			56		2.0 V		"	"		2.7 V	"			"	"	"	"		"	03		"	"
			57			2.0 V	"	"	2.7 V		"			"	"	"	"		"	04		"	"
	I_{OZL}		58				"	"			"		0.4 V	"	"	"	"	2.0 V	"	01		-20	"
			59	2.0 V			"	"			"	0.4 V		"	"	"	"		"	02		"	"
			60		2.0 V		"	"		0.4 V	"			"	"	"	"		"	03		"	"
			61			2.0 V	"	"	0.4 V		"			"	"	"	"		"	04		"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 3/4/ T_C = +25°C	Truth table tests	3014	62	B	A	B	B	B	L	H	GND	L	H	B	C	B	B	A	5.0 V	5/			
		"	63	"	"	"	A	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	64	"	"	"	"	B	"	"	"	"	"	"	"	A	B	"	"	"			
		"	65	"	"	"	A	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	66	A	B	A	B	B	H	L	"	H	L	"	"	B	B	B	"	"			
		"	67	"	"	"	A	A	L	H	"	L	H	"	"	"	"	"	"	"			
		"	68	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"			
		"	69	"	"	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"			
		"	70	"	"	"	B	B	H	L	"	H	L	"	A	"	"	"	"	"			
		"	71	"	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"			
		"	72	"	"	"	A	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	73	"	"	"	A	B	"	"	"	"	"	"	"	A	B	"	"	"			
		"	74	"	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	75	"	A	"	B	B	"	H	"	"	H	"	"	B	B	A	"	"			
		"	76	"	"	"	A	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	77	"	"	"	A	B	"	"	"	"	"	"	"	A	B	"	"	"			
		"	78	"	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	79	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"			
		"	80	B	B	B	B	B	"	"	"	"	"	"	"	B	B	B	"	"			
		"	81	"	"	"	A	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	82	"	"	"	A	B	"	"	"	"	"	"	"	A	B	"	"	"			
		"	83	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 Fig. 4	84				GND	GND			GND		OUT	GND	GND	GND	GND	IN	5.0 V	D1 to Q1	2	50	ns
		"	85	IN			"	"			"	OUT		"	"	"	"		"	D2 to Q2	"	"	"
		"	86		IN		"	"		OUT	"			"	"	"	"		"	D3 to Q3	"	"	"
		"	87			IN	"	"	OUT		"			"	"	"	"		"	D4 to Q4	"	"	"

See footnotes at end of device type 01.

MIL-M-38510/319B

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Subgroup	Symbol	MIL-STD-883-method	Case 2 Cases E,F Test no.	2 1 D2	3 2 D3	4 3 D4	5 4 RB	7 5 RA	8 6 Q4	9 7 Q3	10 8 GND	12 9 Q2	13 10 Q1	14 11 GR	15 12 GW	17 13 WB	18 14 WA	19 15 D1	20 16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHL1}	3003 Fig. 4	88	IN	IN	IN	GND	GND	OUT	OUT	GND	OUT	OUT	GND	GND	GND	GND	IN	5.0 V	D1 to Q1	2	45	ns
		"	89				"	"			"			"	"	"	"		"	D2 to Q2	"	"	"
		"	90				"	"			"			"	"	"	"		"	D3 to Q3	"	"	"
		"	91				"	"			"			"	"	"	"		"	D4 to Q4	"	"	"
	t_{PLH2} 7/	"	92	GND	GND	GND	"	IN	OUT	OUT	"	OUT	OUT	"	"	"	"	GND	"	RA to Q1	"	"	"
		"	93				"	"			"			"	"	"	"		"	RA to Q2	"	"	"
		"	94				"	"			"			"	"	"	"		"	RA to Q3	"	"	"
		"	95				"	"			"			"	"	"	"		"	RA to Q4	"	"	"
	t_{PHL2} 6/	"	96	5.0 V	5.0 V	5.0 V	IN	GND	OUT	OUT	"	OUT	OUT	"	"	5.0 V	"	5.0 V	"	RB to Q1	"	50	"
		"	97				"	"			"			"	"	"	"		"	RB to Q2	"	"	"
		"	98				"	"			"			"	"	"	"		"	RB to Q3	"	"	"
		"	99				"	"			"			"	"	"	"		"	RB to Q4	"	"	"
	t_{PLH3} 6/	"	100	5.0 V	5.0 V	5.0 V	GND	5.0 V	OUT	OUT	"	OUT	OUT	"	IN	GND	IN	5.0 V	"	GW to Q1	"	"	"
		"	101				"	"			"			"	"	"	"		"	GW to Q2	"	"	"
		"	102				"	"			"			"	"	"	"		"	GW to Q3	"	"	"
		"	103				"	"			"			"	"	"	"		"	GW to Q4	"	"	"
	t_{PHL3} 7/	"	104	GND	GND	GND	"	"	OUT	OUT	"	OUT	OUT	"	"	"	"	GND	"	GW to Q1	"	55	"
		"	105				"	"			"			"	"	"	"		"	GW to Q2	"	"	"
		"	106				"	"			"			"	"	"	"		"	GW to Q3	"	"	"
		"	107				"	"			"			"	"	"	"		"	GW to Q4	"	"	"
	t_{PHZ}	"	108	5.0 V	5.0 V	5.0 V	"	GND	OUT	OUT	"	OUT	OUT	IN	GND	"	GND	5.0 V	"	GR to Q1	"	"	"
		"	109				"	"			"			"	"	"	"		"	GR to Q2	"	"	"
		"	110				"	"			"			"	"	"	"		"	GR to Q3	"	"	"
		"	111				"	"			"			"	"	"	"		"	GR to Q4	"	"	"
	t_{PZH}	"	112	5.0 V	5.0 V	5.0 V	"	"	OUT	OUT	"	OUT	OUT	"	"	"	"	5.0 V	"	GR to Q1	"	40	"
		"	113				"	"			"			"	"	"	"		"	GR to Q2	"	"	"
		"	114				"	"			"			"	"	"	"		"	GR to Q3	"	"	"
		"	115				"	"			"			"	"	"	"		"	GR to Q4	"	"	"
	t_{PLZ}	"	116	GND	GND	GND	"	"	OUT	OUT	"	OUT	OUT	"	"	"	"	GND	"	GR to Q1	"	"	"
		"	117				"	"			"			"	"	"	"		"	GR to Q2	"	"	"
		"	118				"	"			"			"	"	"	"		"	GR to Q3	"	"	"
		"	119				"	"			"			"	"	"	"		"	GR to Q4	"	"	"
	t_{PZL}	"	120	GND	GND	GND	"	"	OUT	OUT	"	OUT	OUT	"	"	"	"	GND	"	GR to Q1	"	45	"
		"	121				"	"			"			"	"	"	"		"	GR to Q2	"	"	"
		"	122				"	"			"			"	"	"	"		"	GR to Q3	"	"	"
		"	123				"	"			"			"	"	"	"		"	GR to Q4	"	"	"
10	t_{PLH1} t_{PHL1} t_{PLH2} t_{PHL2} t_{PLH3} t_{PHL3} t_{PHZ} t_{PZH} t_{PLZ} t_{PZL}	"	124	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	"	65	"
		"	125																		"	58	"
		"	126																		"	65	"
		"	127																		"	72	"
		"	128																		"	52	"
		"	129																		"	58	"
		"	130																		"		"
		"	131																		"		"
		"	132																		"		"
		"	133																		"		"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of device type 01.

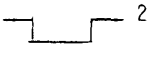
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1/ Min/Max limits (μA) for circuits, unless otherwise specified.

Test	A	B	C	E	F
I_{IL1}	-120/-360	-30/-300	-160/-400 except -130/-360 for test 24	-120/-360 except -0.5/-360 for test 22	-135/-370 except -150/-380 for test 23; -135/-380 for tests 24, 25, 26
I_{IL2}	-36 mA/ -1.08 mA	-90/-900	-320/-1300	-.36 mA/ -1.08 mA	-.48 mA/ -1.20 mA
I_{IL3}	-240/-720	-60/-600	-320/-760	-240/-720	-320/-800

2/ I_{OS} limits for circuits B, C, and E are -15/-100 mA.

3/ Connect a 0.5 kΩ to 2 kΩ resistor from terminals, 6, 7, 9, and 10 to V_{CC} for subgroup 7 tests.

4/ A = 3.0 V min.
B = 0.0 V or GND
C = pulse 2.0 V / 0.7 V  2.0 V

5/ Output voltages shall be:

H > 1.5 V
L < 1.5 V

6/ Load ground into register under test prior to test.

7/ Load one-state into register under test prior to test.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2 Cases E,F Test no.	2 1 D2	3 2 D3	4 3 D4	5 4 RB	7 5 RA	8 6 Q4	9 7 Q3	10 8 GND	12 9 Q2	13 10 Q1	14 11 GR	15 12 GW	17 13 WB	18 14 WA	19 15 D1	20 16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1				0.7 V	0.7 V			GND		4 mA	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	4.5 V	Q1		0.4	V
		"	2	0.7 V			"	"			"	4 mA		"	"	"	"		"	Q2		"	"
		"	3		0.7 V		"	"		4 mA	"			"	"	"	"		"	Q3		"	"
		"	4			0.7 V	"	"	4 mA		"			"	"	"	"		"	Q4		"	"
	I_{CEX}		5				"	"			"		5.5 V	"	"	"	"	2.0 V	"	Q1		20	μA
			6	2.0 V			"	"			"	5.5 V		"	"	"	"		"	Q2		"	"
			7		2.0 V		"	"		5.5 V	"			"	"	"	"		"	Q3		"	"
			8			2.0 V	"	"	5.5 V		"			"	"	"	"		"	Q4		"	"
	V_{IC}		9	-18 mA							"								"	D2		-1.5	V
			10		-18 mA						"								"	D3		"	"
			11			-18 mA					"								"	D4		"	"
			12				-18 mA				"								"	RB		"	"
			13					-18 mA			"								"	RA		"	"
			14								"			-18 mA					"	GR		"	"
			15								"				-18 mA				"	GW		"	"
			16								"					-18 mA			"	WB		"	"
			17								"						-18 mA		"	WA		"	"
			18								"							-18 mA	"	D1		"	"
	I_{IL1}	3009	19	0.4 V							"								5.5 V	D2	1/	1/	μA
		"	20		0.4 V						"								"	D3	"	"	"
		"	21			0.4 V					"								"	D4	"	"	"
		"	22				0.4 V				"								"	RB	"	"	"
		"	23					0.4 V			"								"	RA	"	"	"
		"	24								"					0.4 V			"	WB	"	"	"
		"	25								"						0.4 V		"	WA	"	"	"
		"	26								"							0.4 V	"	D1	"	"	"
	I_{IL2}	"	27								"			0.4 V					"	GR	"	"	"
	I_{IL3}	"	28								"				0.4 V				"	GW	"	"	"
	I_{IH1}	3010	29	2.7 V							"								"	D2		20	"
		"	30		2.7 V						"								"	D3		"	"
		"	31			2.7 V					"								"	D4		"	"
		"	32				2.7 V				"								"	RB		"	"
		"	33					2.7 V			"								"	RA		"	"
		"	34								"					2.7 V			"	WB		"	"
		"	35								"						2.7 V		"	WA		"	"
		"	36								"							2.7 V	"	D1		"	"
	I_{IH2}	"	37								"			2.7 V					"	GR		40	"
	I_{IH3}	"	38								"				2.7 V				"	GW		40	"
	I_{IH4}	"	39	5.5 V							"								"	D2		100	"
		"	40		5.5 V						"								"	D3		"	"
		"	41			5.5 V					"								"	D4		"	"
		"	42				5.5 V				"								"	RB		"	"
		"	43					5.5 V			"								"	RA		"	"
		"	44								"					5.5 V			"	WB		"	"
		"	45								"						5.5 V		"	WA		"	"
		"	46								"							5.5 V	"	D1		"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883-method	Case 2 Cases E,F Test no.	2 1	3 2	4 3	5 4	7 5	8 6	9 7	10 8	12 9	13 10	14 11	15 12	17 13	18 14	19 15	20 16	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IH5}	3010	47								GND			5.5 V					5.5 V	GR		200	µA
	I_{IH6}	"	48								"				5.5 V				"	GW		"	"
	I_{CC}	3005	49	4.5 V	4.5 V	4.5 V	GND	GND			"			4.5 V	4.5 V	GND	GND	4.5 V	"	V_{CC}		40	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 2/3/ T_C = +25°C	Truth table tests	3014	50	B	A	B	B	B	L	H	GND	L	H	B	C	B	B	A	5.0 V	4/			
		"	51	"	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	52	"	"	"	A	B	"	"	"	"	"	"	"	A	B	"	"	"			
		"	53	"	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	54	A	B	A	B	B	H	L	"	H	L	"	"	B	B	B	"	"			
		"	55	"	"	"	"	A	L	H	"	L	H	"	"	"	"	"	"	"			
		"	56	"	"	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"			
		"	57	"	"	B	"	A	"	"	"	"	"	"	A	"	"	"	"	"			
		"	58	"	"	"	B	B	H	L	"	H	L	"	"	A	"	"	"	"			
		"	59	"	"	"	"	"	"	H	"	"	H	A	C	B	"	"	"	"			
		"	60	"	"	A	"	A	"	L	"	"	L	B	"	"	A	"	"	"			
		"	61	"	"	"	A	B	"	"	"	"	"	"	"	A	B	"	"	"			
		"	62	"	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	63	"	A	"	B	B	"	H	"	"	H	A	"	B	B	A	"	"			
		"	64	"	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	65	"	"	"	A	B	"	"	"	"	"	"	"	A	B	"	"	"			
		"	66	"	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	67	B	B	B	B	B	"	"	"	"	"	9	A	B	B	B	"	"			
		"	68	"	"	"	"	A	"	"	"	"	"	"	"	"	A	"	"	"			
		"	69	"	"	"	A	B	"	"	"	"	"	"	"	A	B	"	"	"			
		"	70	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 Fig. 4	71				GND	GND			GND		OUT	GND	GND	GND	GND	IN	5.0 V	D1 to Q1	2	50	ns
			72	IN			"	"			"	OUT		"	"	"	"		"	D2 to Q2	"	"	"
			73		IN		"	"		OUT	"	"		"	"	"	"		"	D3 to Q3	"	"	"
			74			IN	"	"	OUT	"	"	"		"	"	"	"		"	D4 to Q4	"	"	"
	t_{PHL1}	"	75				GND	GND			"		OUT	"	"	"	"	IN	"	D1 to Q1	"	40	"
		"	76	IN			"	"			"	OUT		"	"	"	"		"	D2 to Q2	"	"	"
		"	77		IN		"	"		OUT	"	"		"	"	"	"		"	D3 to Q3	"	"	"
		"	78			IN	"	"	OUT	"	"	"		"	"	"	"		"	D4 to Q4	"	"	"
	t_{PLH2} 5/	"	79				"	IN			"		OUT	"	"	"	5.0 V	5.0 V	"	RA to Q1	"	45	"
		"	80	5.0 V			"	"			"	OUT		"	"	"	"		"	RA to Q2	"	"	"
		"	81		5.0 V		"	"		OUT	"	"		"	"	"	"		"	RA to Q3	"	"	"
		"	82			5.0 V	"	"	OUT	"	"	"		"	"	"	"		"	RA to Q4	"	"	"
	t_{PHL2} 5/	"	83				IN	GND			"		OUT	"	"	5.0 V	GND	5.0 V	"	RB to Q1	"	"	"
		"	84	5.0 V			"	"			"	OUT		"	"	"	"		"	RB to Q2	"	"	"
		"	85		5.0 V		"	"		OUT	"	"		"	"	"	"		"	RB to Q3	"	"	"
		"	86			5.0 V	"	"	OUT	"	"	"		"	"	"	"		"	RB to Q4	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case & Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
9 T_C = +25°C	t_{PLH3} 5/	3004 Fig. 4	87				GND	5.0 V					OUT	GND	IN	GND	IN	5.0 V	5.0 V	GW to Q1	2	50	ns
		"	88	5.0 V			"	"				"	"	"	"	"	"		"	GW to Q2	"	"	"
		"	89		5.0 V		"	"		OUT		"		"	"	"	"		"	GW to Q3	"	"	"
		"	90			5.0 V	"	"	OUT			"		"	"	"	"		"	GW to Q4	"	"	"
	t_{PHL3} 6/	"	91				"	"				"	OUT	"	"	"	"	GND	"	GW to Q1	"	45	"
		"	92	GND			"	"				"	"	"	"	"	"		"	GW to Q2	"	"	"
		"	93		GND		"	"		OUT		"		"	"	"	"		"	GW to Q3	"	"	"
		"	94			GND	"	"	OUT			"		"	"	"	"		"	GW to Q4	"	"	"
	t_{PLH4}	"	95				"	GND				"	OUT	OUT	IN	GND	"	GND	GND	"	GR to Q1	"	40	"
		"	96	GND			"	"				"	"	"	"	"	"		"	GR to Q2	"	"	"
		"	97		GND		"	"		OUT		"		"	"	"	"		"	GR to Q3	"	"	"
		"	98			GND	"	"	OUT			"		"	"	"	"		"	GR to Q4	"	"	"
	t_{PHL4}	"	99				"	"				"	OUT	OUT	"	"	"	GND	"	GR to Q1	"	"	"
		"	100	GND			"	"				"	"	"	"	"	"		"	GR to Q2	"	"	"
		"	101		GND		"	"		OUT		"		"	"	"	"		"	GR to Q3	"	"	"
		"	102			GND	"	"	OUT			"		"	"	"	"		"	GR to Q4	"	"	"
10	t_{PLH1}	"	103	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																	"	65	"
	t_{PHL1}	"	104																		"	52	"
	t_{PLH2}	"	105																		"	58	"
	t_{PHL2}	"	106																		"	"	"
	t_{PLH3}	"	107																		"	65	"
	t_{PHL3}	"	108																		"	58	"
	t_{PLH4}	"	109																		"	46	"
	t_{PHL4}	"	110																		"	"	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Min/Max limits (μA) for circuits, unless otherwise specified.

Test	A	B	C	E	F
I_{IL1}	-120/-360	-30/-300	-190/-420 except -160/-460 for test 24	-120/-360 except -0.5/-360 for tests 22 and 24	-135/-370 except -150/-380 for test 23
I_{IL2}	-240/-720	-90/-900	-380/-840	-240/-720	-320/-800
I_{IL3}	-240/-720	-60/-600	-380/-840	-240/-720	-320/-800

2/ Connect a 0.5 kΩ to 2 kΩ resistor from terminals, 6, 7, 9, and 10 to V_{CC} for subgroup 7 tests.

3/ A = 3.0 V minimum, B = 0.0 V or GND, C = pulse 2.0 V ⊓ 2.0 V / 0.7 V

4/ Output voltages shall be:

H > 1.5 V.
L < 1.5 V.

5/ Load ground into register under test prior to test.

6/ Load one-state into register under test prior to test.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical tests shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.6 Inclusion with other device types. For qualification and quality conformance purposes, either device type as covered by this specification may be treated as though it were included with the other device type. For example, if a manufacturer is already part I qualified for MIL-M-38510/31901 and if the device MIL-M-38510/31902 is designed and manufactured identically, (same die, same process, same screening,) in all respects except electrical testing, then device type 31902 may, when authorized by the qualifying activity, be part I qualified by conducting only group A electrical test and submitting data in accordance wth MIL-M-38510, appendix D (i.e., groups B, C, and D tests are not required).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
I_{IN}	Current flowing into an input terminal.
V_{IC}	Input clamp voltage.
V_{IN}	Voltage level at an input terminal.
I_{OZL}	Output current in the high-impedance mode with the output voltage low.

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I_{OZH} - - - - - - - - - - - - - - Output current in the high-impedance mode with the output voltage high.

t_{PHZ} - - - - - - - - - - - - - - Output disable time from high level - The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high-impedance (off) state.

t_{PLZ} - - - - - - - - - - - - - - Output disable time from low level - The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.

t_{PZH} - - - - - - - - - - - - - - Output enable time to high level - The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined high level.

t_{PZL} - - - - - - - - - - - - - - Output enable time to low level - The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS670
02	54LS170

6.6 Manufacturer's designation. Manufacturers' included in this specification are designated as shown in table IV herein.

TABLE IV. Substitutability and manufacturers' designation.

Device type	Manufacturer				
	Circuit A	Circuit B	Circuit C	Circuit E	Circuit F
	Motorola Inc.	Signetics Corp.	National Semi-conductor Corp.	Fairchild Semiconductor	Motorola, Inc.
01	X	X	X	X	X
02	X	X	X	X	X

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/319B	~~Microcircuits, digital, low power Schottky TTL,~~ 4- by-4 register file, monolithic silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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